POSTASSIST LK · J26-IT-372 · INDIVIDUAL COMPONENT

My Research Work – Complete Record

Smart Registered Mail Intake and Tracking: everything built, trained, measured and written so far, what the results show, what changed from the proposal, and what is still left. Status on 20 October 2026.

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HOW THIS RECORD WAS MADE

Every number comes from the repository on 20 October 2026: the git history, the model reports in `ml/reports/`, the backend test run of that day (71 of 71 component tests passed), and the development database (read only). Results from **synthetic** data are labelled as such and kept apart from results on **real** data.

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Branch
feature/smart_registered_mail_intake_&_tracking
Status: 20 October 2026

PART 1

At a glance

The whole component in one page.

I have built a complete, working system that **accepts registered letters at a post-office counter** and **tracks them until delivery**: a camera and trained models read the envelope address (Sinhala, Tamil, English, printed or handwritten), my IoT scale weighs the letter, the official 2026 tariff gives the price, a barcode/QR label is printed, every later scan is checked for permission, valid order and GPS evidence, phones keep working offline, and the sender gets reliable e-mail updates and a private tracking page.

6
parts built: backend, web app, 2 phone apps, IoT device, ML pipeline

3
models trained: script ID (CNN), address parser (CRF), handwriting reader (CRNN)

70
API endpoints (FastAPI)

71 / 71
backend tests of my component pass (20 Oct)

15 / 18
acceptance tests pass; 2 implemented, 1 to run on a phone

143
real weighing trials recorded on the IoT scale

9
public datasets used (with licences)

≈ 33,000
lines of code (backend, web, mobile, firmware, ML)

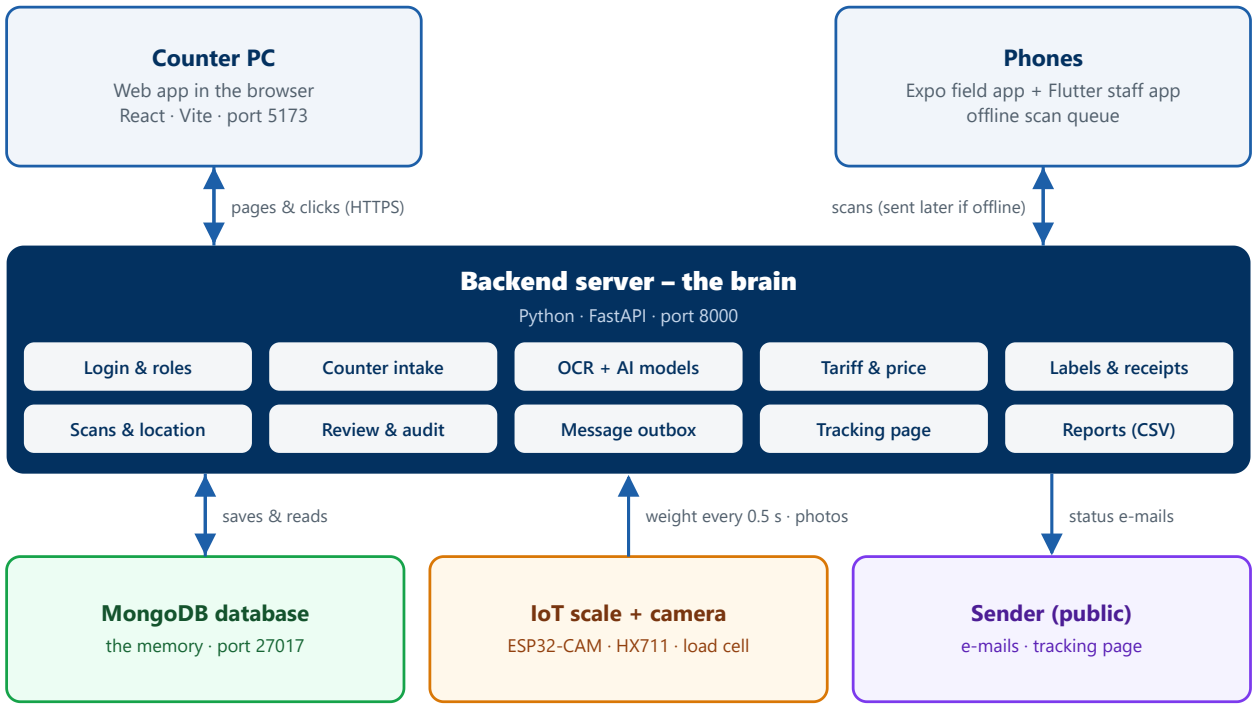


Figure 1 – The parts of the component and how they connect.

DONE

- All software parts built and working together
- Three models trained and evaluated on held-out data
- IoT scale + camera built, programmed, connected; real letters weighed
- Location policies L0/L1/L2 built and simulated
- Offline sync and message outbox proven by automated tests
- Requirements, setup, hardware and user guides written

STILL TO DO

- Full 10 × 10 weighing trials (5–500 g) and the uncertainty U
- Field location trials and usability sessions with real users
- Consented real-envelope test set; Tamil handwriting data
- Tariff approval; ethics and Department of Posts approval
- Flash firmware 1.1.1; test the Flutter app on a phone
- Commit the recent work (UI redesign, Flutter app, guides)

PART 2

Problem and objectives

2.1 The problem

Between accepting a domestic registered letter at the counter and giving the sender a credible delivery update, dependable information is lost in four places: **retyping** the envelope address, **weight and tariff transcription errors**, status updates **without contextual evidence**, and **inconsistent communication** when the network or a messaging service fails.

2.2 Scope

Domestic registered letters of **5–500 g**, one pilot counter, a few approved offices, one delivery team, **synthetic mail** until approvals. Users: counter officer, hub officer, postman, postmaster, administrator, sender, recipient, and the IoT device.

2.3 The four research objectives

Objective	Target
SO1 Assisted capture	≥ 20 % lower median intake time than manual entry; ≥ 98 % confirmed required-field exact match; handwriting failures reported.
SO2 IoT weighing and tariff	MAE ≤ 1 g over 5–500 g; 100 % tariff agreement including band edges; boundary uncertainty never silently rounded.

Objective	Target
SO3 Location-aware scans	Compare L0 / L1 / L2; ≤ 5 % false acceptance of clearly-outside scans; report rejection and uncertainty rates.
SO4 Offline sync and messaging	Zero duplicate events in 1,000 replays; zero messages from rolled-back or pending events; all saved offline events recovered.

2.4 One letter's journey through the system

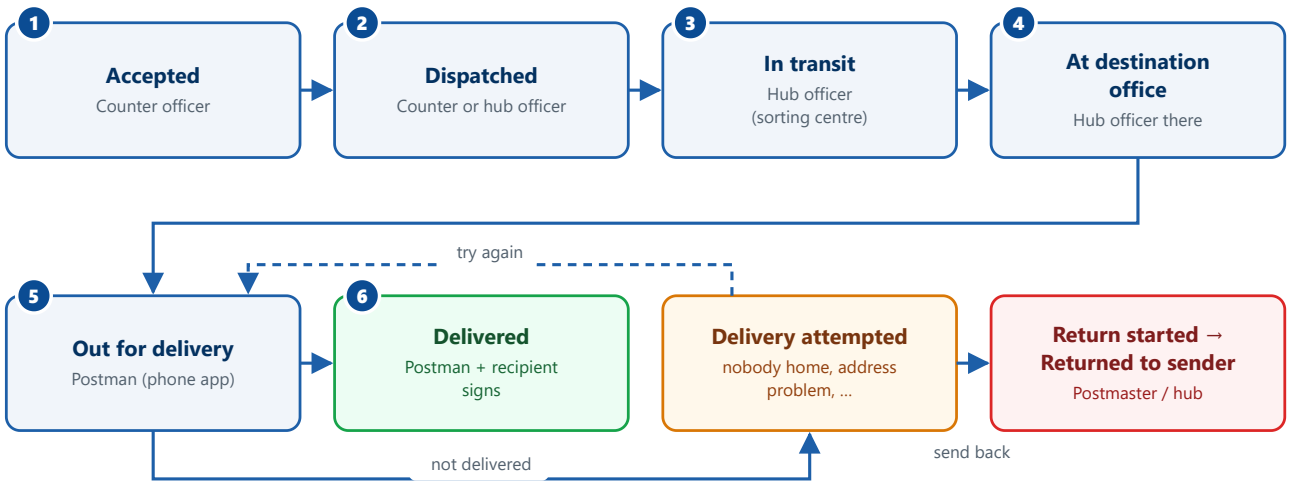


Figure 2 – The statuses a letter moves through.

PART 3

Timeline – what was done when

From the git history and the work sessions.

Date (2026)	Work
16 Aug	Group repository and project structure created.
Sept	Project proposal (component: Smart Registered Mail Intake and Tracking).
1 Oct	Component structure; public reference datasets, curated resources (trilingual vocabulary, name list, 2026 tariff) and data-collection templates. **Address CRF trained.** Requirements specification v1.0.
2 Oct	**FastAPI backend**: intake, tariff, scans, review, notifications. **Script-ID CNN trained** (3 seeds).
3 Oct	**ESP32-CAM firmware 1.0** (scale + envelope camera); React web app scaffold with routing, API client, login and item views.
5 Oct	Dataset builders, CRF and script-ID training code, evaluation scripts and trained models committed.

Date (2026)	Work
11 Oct	Real hardware connected through the **device bridge** (board ran a pull sketch); first real letter weighed (11.4 g) and accepted; weighing trials started. **Handwriting recogniser trained** (40 epochs on the RTX 2050). Login page.
14 Oct	Handwriting reader added to the P1 pipeline; integration test flow and setup guide; counter pages, weight panel, theme; PP1 slides fact-checked (corrected copy in Documents).
18 Oct	**Firmware 1.1** (sharper photos, focus helper, stable HX711 reads, on-device network settings) flashed and running; backend **starts MongoDB automatically**; web app pages (intake wizard, label print, items, tracking, IoT, review, messages, offices, audit) and the Expo field app committed; teammate's **KPI desk merged** into one shared backend; **firmware 1.1.1** written (calibration safeguards, not yet flashed); "Delete" for unfinished drafts.
18–19 Oct	Plain-language component guides (English, Sinhala) and the IoT guide; web pages restyled (grouped sidebar, intake stepper, plain status labels).
19–20 Oct	**Flutter staff app** for all five roles in Sinhala, Tamil and English; release APK built; 32 tests passed.
20 Oct	Full backend test run: 71 of 71 component tests pass. This record.

The last three rows are not committed to git yet (Part 12).

PART 4

What I built

Each part, what it does, and where it is.

4.1 Backend – FastAPI + MongoDB (`backend/` , ≈ 5,200 lines)

- **11 routers, 70 endpoints**, documented automatically at `/docs`: login, intake, items, scans, review and audit, notifications, devices, IoT device API, public tracking, administration, reports.
- **Counter intake**: drafts that save themselves; image upload, webcam or scale capture; quality check; OCR (B1, P1, handwriting); address parsing with confidence; town/postcode check; field confirmation with correction statistics; scale pairing rules; audited manual weight; quote; idempotent acceptance; tracking IDs with check character; Code 128 + QR labels.
- **Tariff engine**: versioned 2026 tariff, validation (gaps, overlaps, one tariff in force), itemised quote, boundary-uncertainty rule ("reweigh required").
- **Scan decision pipeline** (8 checks): item, replay, role and office permission, valid transition / late history, delivery acknowledgement, clock and mock-GPS, location assessment, atomic commit.
- **Location policies** L0 / L1 / L2 (probability inside the zone from the GPS accuracy).
- **Transactional outbox** with a background worker: de-duplication, bounded back-off, honest "ambiguous" outcome; e-mail by file, SMTP or Resend; English, Sinhala and Tamil templates.
- **Security**: PBKDF2 passwords, 12-hour JWT, role and object-level rules, device keys, audit log, no personal data in codes, URLs or exports.
- **Reports**: three CSV exports for the evaluation (no names or addresses) and a dashboard summary.

- **Operations**: demo seed (6 offices, 10 accounts, tariff, policy, device); MongoDB auto-start; one-click start scripts.

4.2 Web app – React (`frontend/src/`, ≈ 3,000 lines of my pages)

Login, Dashboard, Counter intake (draft list with delete), the 5-step intake screen, Label & receipt with scan-back verification, Registered items, Item history, Record a scan, Review queue, Sender messages, Audit log, Scale & camera, Offices & zones, Reports & settings, and the public tracking page. Keyboard accessible, text-plus-colour badges, Sinhala/Tamil fonts.

4.3 Phone apps (`mobile_app/`)

- **Expo React Native field app** (≈ 650 lines): sign-in, scan Code 128 / QR, record the status with GPS evidence, signature / name / delivery-sheet acknowledgement, **offline SQLite queue** with automatic ordered upload.
- **Flutter staff app** (`mobile_app/flutter_app/`, ≈ 18,500 lines): all five roles – counter intake in 5 steps, label and verification, hub and postman scans (offline), review queue, summary, audit, offices, public tracking – in **Sinhala, Tamil and English**. `flutter analyze` clean, 32 tests passed including a contract test against a live backend, release APK built. Not yet run on a real phone.

4.4 IoT scale and camera (`iot/`, ≈ 800 lines)

ESP32-CAM (OV3660) + 1 kg load cell + HX711 + level shifter. Firmware in Arduino C++: own HX711 read with slow clock pulses, 1-second trimmed-mean window, stability (≤ 1 g spread), placement numbering, spike rejection, zero tracking, calibration in flash, reports every 0.5 s with a device key, commands (tare, calibrate, capture, identify, reboot) carried back in the reply, best-of-3 photo at 2048 × 1536, its own web page with focus helper and Wi-Fi settings, set-up network "PostAssist-Scale". Plus a **device simulator** and a **bridge** for older pull-only boards.

4.5 Machine learning (`ml/`, ≈ 3,600 lines; models in `backend/models/`)

- **Script identification CNN** (0.46 M parameters, ONNX, temperature-scaled).
- **Address-field CRF** with gazetteer and trilingual-vocabulary features, calibrated confidence and a frozen review threshold.
- **Handwriting recogniser** CRNN + CTC (4.5 M parameters, 195 characters, ONNX), with photo clean-up (ink image), line finding and digit-aware decoding.
- Dataset builders, 360 synthetic envelope images, OCR benchmark, weighing analysis (uncertainty U), location-policy simulation, one-command rebuild (`ml/run_all.sh`).

4.6 Tests

71 backend tests: unit tests (tariff, tracking IDs, location, address, guard), end-to-end workflow tests for acceptance scenarios T03–T17 including **1,000 scan replays**, handwriting pre-processing, the device bridge, and the shared server with the KPI desk. All passed on 20 October (about 70 s). Flutter: 32 tests.

PART 5

Objectives – evidence status

For each objective: what the system can do, what has been measured, and what is missing.

Obj.	Built	Evidence so far	Missing
SO1	B0 / B1 / P1 configurations; OCR + handwriting + CRF; correction statistics and timings per draft; intake CSV export	CRF 99.4 % required-field exact match on unseen towns; handwriting CER 18.1 % (Sinhala) / 9.2 % (English) on real held-out lines; script ID 98.6 %	Timed B0 vs P1 sessions with users; real consented envelopes; whole-envelope OCR is weak on synthetic envelopes (6.4)
SO2	IoT scale, pairing rules, tariff engine, boundary rule, trials page, analysis script	143 real trials, MAE 0.18 g (10–21 g only); tariff unit tests incl. band edges and boundary rule pass	The 5–500 g, 10 × 10 protocol; run the analysis; enter U; certified reference masses
SO3	L0 / L1 / L2, reference zones with provenance, review queue, events CSV	Simulation (20,000 scans per scenario): false acceptance L0 100 %, L1 0 %, L2 0 %; acceptance tests T07–T10 pass	Controlled field trials with 2–3 phones at the 6 areas
SO4	Offline queue (Expo and Flutter), idempotent server, transactional outbox	1,000 replays → 1 event, 1 message (T12); no message from held events (T14); provider failures (T15) – all pass	Airplane-mode restart test on a phone (T11)

PART 6

Results so far

Real data and synthetic data are reported separately.

6.1 Script identification (CNN) – unseen fonts

Measure	Result
Test accuracy, shipped model (6 fonts never seen, held-out text)	**98.6 %**
Across 3 training seeds	97.5 % ± 1.4 %
Printed lines / handwriting-style lines	99.5 % / 94.1 %
Handwriting-style lines across seeds (the weak spot)	87.5 % ± 7.9 %
Calibration error (ECE) before → after temperature scaling	0.067 → 0.012
Speed on CPU (ONNX)	2–7 ms per line

Rendered lines, not photographs; no Sinhala handwriting font existed for this model.

6.2 Address-field parser (CRF) – towns never seen in training

Measure (test)	CRF	Rule baseline
Token accuracy	**99.7 %**	74.6 %
Required-field exact match	**99.4 %**	57.5 %
Whole-address exact match	**98.3 %**	19.6 %
Real OpenStreetMap-based addresses (required-field EM)	97.6 %	–
With OCR-style noise (required-field EM)	98.7 %	–
By script: Sinhala / Tamil / English / mixed	99.7 / 99.8 / 98.9 / 99.9 %	61 / 62 / 52 / 61 %

Confidence is calibrated (ECE 0.0024). The review threshold **0.974** was chosen on validation and frozen: on test it flags 1.8 % of fields, the un-flagged fields are 99.8 % correct, and it catches 68 % of the errors. Data: 24,245 / 4,878 / 5,337 addresses (train / val / test), grouped by postal town.

6.3 Handwriting recogniser (CRNN + CTC)

Held-out test set	Lines	Model CER	Model WER	Tesseract CER
Real Sinhala handwriting (SinOCR test)	227	**18.1 %**	55.5 %	91.1 %
Real English handwriting (IAM test sample)	400	**9.2 %**	31.1 %	55.4 %
Tamil, fonts never seen (synthetic)	300	0.4 %	2.4 %	–
English address text, unseen handwriting-style font (synthetic)	300	2.0 %	13.1 %	–
Your own two phone photos (English / Sinhala)	2	7 % / 21 %	–	–

Tamil handwriting is untested (no real data). About every second handwritten word still has an error, so the officer always confirms.

6.4 Whole-envelope OCR benchmark (synthetic envelopes, 67 test images)

Configuration	Median CER	Required-field exact match	Median time
B1 – Tesseract, all three packs, rules	0.96	16.6 %	522 ms
P1 – assisted pipeline	1.00	16.9 %	485 ms
P1 + handwriting model	1.00	16.6 %	758 ms

HONEST FINDING

On whole, **uncropped** synthetic envelope images every configuration is at the floor, so this benchmark cannot yet show P1 beating B1. The models work on clean lines and text (6.1–6.3); the weak link is finding the address block on a full envelope. In the app the officer crops the address first. For SO1, evaluate on cropped, consented real envelopes, and consider automatic address-block detection.

6.5 Location policy simulation (20,000 scans per scenario)

Policy	False acceptance, clearly outside (target ≤ 5 %)	Valid scans held as mismatch	Valid scans flagged / held	Stale fix reported consistent
L0 status only	100 %	0 %	0 %	100 %
L1 fixed radius	0 %	6.9 %	6.9 %	90.4 %
L2 uncertainty-aware	**0 %**	**1.7 %**	18.7 %	**0 %**

Simulation only – confirm with field trials before freezing thresholds.

6.6 Real weighing trials on the IoT scale

Reference (g)	Trials	Bias (g)	SD (g)	Max |error| (g)
10.00	29	−0.03	0.02	0.06
12.00	12	+1.24	0.04	1.33
14.28	7	+0.07	0.02	0.08
15.00	34	+0.02	0.11	0.29
16.56	12	−0.01	0.01	0.02
18.84	17	−0.03	0.05	0.10
20.00	18	+0.05	0.01	0.06
20.94	14	+0.34	0.09	0.52
All 143 trials (11–18 Oct, 8 calibrations)	143		**MAE 0.18 g**	1.33

WHAT THIS DOES AND DOES NOT SHOW

It is promising (MAE below the 1 g target), but it covers only **10–21 g**, spreads over 8 calibrations and uses references whose own accuracy is not documented (the 12 g reference is consistently 1.2 g off – check that object). It is **not yet** the planned 5–500 g, 10 masses × 10 repeats protocol, and `weighing_analysis.py` has not been run (`ml/reports/weighing/` is empty).

6.7 Real end-to-end use (development database)

11 registered items (8 accepted, 2 dispatched, 1 delivered), 18 recorded events, 87 drafts; 2 items weighed by the IoT scale, including a real letter of 11.4 g on 11 October.

6.8 Acceptance tests

Test	Scenario	Status
T01, T02	Clear printed envelope; blurred handwritten Sinhala/Tamil	implemented

Test	Scenario	Status
T03–T06	Reused weight refused; boundary → reweigh; repeated accept → same item; reprint audited	pass
T07–T10	Unauthorised worker denied; outside GPS held; no/stale GPS by policy; delivered needs acknowledgement	pass
T11	Offline save survives an app restart	run on a phone
T12–T17	1,000 replays; late scan history only; no message from held events; provider timeout; URL tampering; return path	pass
T18	Map / geocoder unavailable	pass by design

PART 7

Datasets used

All public or generated; no real customer mail.

Dataset	Content	Licence	Used for
Sri Lanka provinces-districts-cities (Madurapperuma)	2,155 postal towns in EN/SI/TA with postcodes	MIT	Gazetteer, synthetic addresses
Deepparse worldwide addresses (lk)	19,292 OSM-derived Sri Lankan addresses with tags	CC BY 4.0	Real part of the CRF corpus
OpenStreetMap (Overpass)	797 post offices; 358 streets with SI/TA names	ODbL	Office points, vocabulary
Wikidata	Sri Lankan given and family names	CC0	Name vocabulary
Google Fonts (27)	Sinhala, Tamil, Latin, handwriting-style	OFL	Rendering lines and envelopes
Department of Posts notice, 9 Feb 2026	Domestic postage revision	public notice	Versioned tariff
SinOCR – handwritten	1,135 real Sinhala lines (908 / 227)	confirm & cite	Handwriting train/test
Sinhala characters, 454 classes	108,933 character images, no label file (270 labelled by eye)	confirm source	Synthetic Sinhala lines
IAM handwriting – lines	10,373 real English lines	non-commercial research	Handwriting train/test

Generated: 34,460 labelled addresses (24,245 / 4,878 / 5,337), 54,000 rendered script lines, 360 synthetic envelope images. **Templates ready for your own data:** envelope annotation, weighing trials, location trials, usability task log (all still empty).

PART 8

Hardware

Item	Price (LKR)
ESP32-CAM development board with OV3660 3 MP camera	2,750
1 kg bar load cell	450
HX711 24-bit ADC module	160
Wooden load-cell bracket	600
4-channel level shifter (3.3 V ↔ 5 V)	100
Jumper wires	160
Shipping	580
Total (within the Rs 10,500 proposal allowance)	**4,800**

Also used: ESP32-CAM-MB programmer, breadboard, USB breakout and 5 V / 2 A charger, camera stand and lamp; laptop with RTX 2050 (4 GB) for training; iPhone hotspot as the lab network.

PART 9

Documents produced

File (Documents/ unless noted)	What it is
`Requirements_Specification.md`	Functional, non-functional, hardware, business rules, ML requirements, traceability, acceptance tests.
`Setup_and_Run_Guide.md`	Install, run, demo script, experiments.
`IoT_Hardware_Setup_Guide.md`	Step-by-step build, wiring, flashing, calibration, troubleshooting, protocol.
`IoT_Integration_Test_Flow.md`	How to test the device with the system.
`Flutter_Mobile_App_Guide.md`	Setting up and running the Flutter app.
`PP1_Individual_IT23324138_verified.pptx`	PP1 slides, fact-checked (feedback section still to fill with real feedback).
`PostAssist_Folder_Structure_EN_SI.pdf`	Folder structure guide, English and Sinhala.
`PostAssist_Component_Guide_EN.pdf · _SI.pdf`	The whole component explained simply (English and Sinhala).
`PostAssist_IoT_Scale_Guide_EN.pdf`	Everything about the IoT part and its programming.

File (Documents/ unless noted)	What it is
`ml/README.md`, `ml/reports/*/REPORT.md`	Training protocol and evaluation report of every model.
`Datasets/README.md`, `raw/MANIFEST.json`	Every dataset, licence, citation and checksum.

PART 10

Changes from the proposal

Be ready to explain each one.

Proposal	Now	Why
PostgreSQL	**MongoDB**	Each letter is one document that holds its events and messages, so every change is one atomic write – the outbox and idempotency need no multi-table transactions. The group stack is Python + React + MongoDB.
Flutter mobile app	**Expo React Native** field app, and since 19 Oct also a **Flutter** staff app	The field app was built first in React Native; the Flutter app now covers all roles in three languages. Decide which one to present and make the PP1 slide match.
LLM normalisation	**Optional and off** (Gemma 3 4B via Ollama) with a guard	Not needed for accuracy; a guard rejects invented digits, names or script changes.
Tesseract for all OCR	**Own handwriting model** in the P1 pipeline	Tesseract's CER on handwriting was 91 % (Sinhala) and 56 % (English).
HX711 library	**Own HX711 read** (firmware 1.1)	The library's fast clock read wrong bits through the level shifter.
Separate registered-mail server	**One backend** serving this component and the teammate's KPI desk	Merged on 18 Oct; the desk's routes are registered last so they cannot hide ours.

PART 11

Problems faced and how I solved them

Problem	Solution
Tesseract cannot read handwriting	Trained a CRNN-CTC handwriting reader on real Sinhala (SinOCR) and English (IAM) lines plus synthetic lines from real handwritten characters; “auto” lets the more confident reader win.
Sinhala shaping (conjuncts, ZWJ) broke rendering and text	HarfBuzz-shaped rendering; ZWJ handling in decoding; repaired broken conjuncts in the corpus.
Training on a 4 GB GPU slowed several-fold	Capped GPU memory at 85 % and padded batch widths to multiples of 64 (≈ 95 s per epoch).
Address parser over-trusting one source	Grouped splits by postal town so test towns are never seen; calibrated confidence; review threshold frozen on validation.
HX711 readings jumped by fixed amounts	Own bit-banged read with slow 4 µs pulses in a critical section; spike rejection; trimmed mean.
A bad stored calibration locked the scale (never stable)	Firmware 1.1.1 discards impossible factors, adds reset calibration and force tare.
Blurry envelope photos	Focus helper for 30 cm, sharpening on, noise reduction off, best-of-3 burst, warm-up frames.
The board first ran a pull-only sketch	Device bridge on the laptop, then firmware 1.1 that pushes by itself.
Previous letter’s weight could be reused	Placement numbers in firmware + unique placement key in MongoDB.
Price near a band edge is uncertain	Boundary rule: weight ± U across a step → “reweigh required”.
GPS accuracy varies a lot	L2 policy: probability inside the zone from the reported accuracy.
Offline retries could duplicate scans and e-mails	One-time event IDs, idempotent server, transactional outbox with de-duplication keys.
MongoDB stopped when a window closed	The backend starts a detached MongoDB itself when none is running.
Teammate’s catch-all route hid `/api/health` after the merge	Registered the KPI desk last, plus a test that checks the order.
Arduino libraries invisible (non-ASCII sketchbook path)	Moved libraries to a plain path.

PART 12

What is still open

Known gaps and risks, so nothing surprises you at the panel.

12.1 Research evidence not yet collected

- Full 10 × 10 weighing trials over 5–500 g with documented reference masses; run `weighing_analysis.py`; enter U.
- Controlled location field trials (≈ 120 scans at 6 areas, 2–3 phones) for SO3.
- Usability sessions (12–16 participants, task completion, SUS ≥ 70) and timed B0 vs P1 intake for SO1.
- **User feedback on the prototype** – none collected yet; the PP1 slide has empty slots for it. Never present invented feedback.
- Consented real-envelope set (360 images) and real Tamil handwriting.
- Airplane-mode restart test on a phone (T11).

12.2 Approvals

- Ethics committee and Department of Posts approval before any real mail or staff evaluation.
- Tariff marked “transcribed – research use” until an authorised officer approves it.
- Licence and citation of SinOCR and the 454-class character set still to confirm.

12.3 Technical to-dos

- **Commit** the recent work: web redesign (+ `IntakeProgress.jsx`), Flutter app, guides, README changes.
- **Flash firmware 1.1.1**; reset, tare, calibrate; focus the lens.
- Empty `BRIDGE_DEVICE_URL` in `backend/.env` (currently set, so the bridge and the board both report).
- Remove the real hotspot password from `iot/postassist_scale_cam/config.example.h` (it is on GitHub) and change the hotspot password.
- Run the Flutter app on a real phone; check the redesigned web pages in a browser.
- Address-block detection for whole-envelope OCR (6.4).
- Teammate’s test `test_forecast_uses_the_saved_model` fails on Python 3.14 (scikit-learn 1.6.1 model) – theirs to fix.

12.4 Not built (optional features)

Recipient OTP acknowledgement, push notifications, map view of scans, real SMS gateway.

PART 13

Next steps

13.1 Before PP1

1. Decide React Native or Flutter for the demo and make the slides match.
2. Get short feedback from 2–3 real people (e.g. a post-office officer) and fill the slide's feedback slots honestly.
3. Flash 1.1.1, turn off the bridge, calibrate and focus – rehearse the counter flow end to end.
4. Commit and push the recent work.
5. Prepare the honest limitations: synthetic data, Tamil handwriting untested, whole-envelope OCR, DIY scale not legal for trade, GPS is evidence not proof.

13.2 After PP1

1. Weighing protocol and uncertainty U (SO2).
2. Location field trials (SO3) and the phone restart test (SO4, T11).
3. Ethics approval → consented envelopes → SO1 timing study and usability (SUS).
4. Improve whole-envelope OCR (address-block detection) and evaluate on the consented set.
5. Tamil handwriting data (e.g. IIIT-INDIC-HW-WORDS, 5.1 GB – needs approval to download).
6. Write the evaluation chapter from the CSV exports and reports.

PART 14

Where everything is

Folder	Contents
`backend/app/`	The API: routers, services (OCR, address, tariff, weighing, scans, location, notifications), security, seed.
`backend/models/`	address_crf, script_id.onnx, htr.onnx (+ metadata).
`backend/tests/`	71 tests of the component.
`backend/tools/`	start_all.bat / .ps1, device simulator, device bridge.
`frontend/src/`	Web app pages and components (KPI desk pages belong to the teammate).
`mobile_app/` · `mobile_app/flutter_app/`	Expo field app · Flutter staff app.
`iot/postassist_scale_cam/`	Firmware 1.1.1 and its web page.
`ml/`	Dataset builders, training, evaluation, reports.
`Datasets/`	Raw, curated and processed data, templates.
`Documents/`	Requirements, guides, slides, PDFs.
`%USERPROFILE%\PostAssist\`	MongoDB program and data; handwriting datasets and training cache (outside the repository).